Ch 7.4: Cubic splines

Lecture 24 - CMSE 381

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~~Weds~~, Nov 6, 2023

Monday

Announcements

Last time:

- 7.2 Step functions
- 7.3 Basis functions

This lecture:

- 7.4 Cubic splines

Announcements:

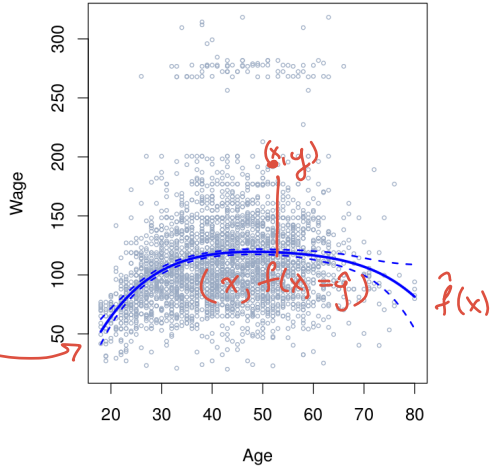
- Homework # 6 is now due Wednesday

Section 1

Last time

Polynomial regression

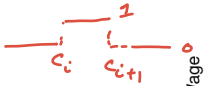
$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$



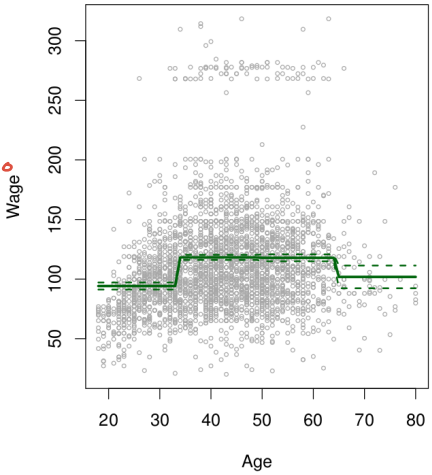
Step function regression

Knots: $c_1, \ldots, c_K$

$$
\begin{aligned}
C_0(X) &= I(X < c_1), \\
C_1(X) &= I(c_1 \leq X < c_2), \\
C_2(X) &= I(c_2 \leq X < c_3), \\
&\vdots \\
C_{K-1}(X) &= I(c_{K-1} \leq X < c_K), \\
C_K(X) &= I(c_K \leq X),
\end{aligned}
$$

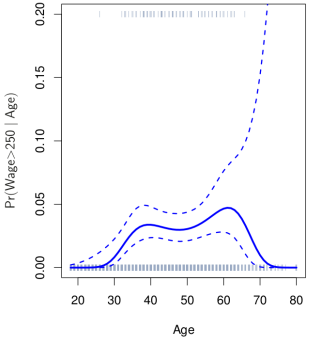


$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i) + \varepsilon_i$$



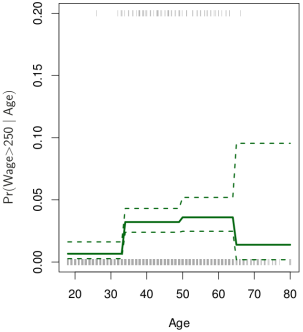
Classification version

$$\frac{\exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}{1 + \exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}$$



$$\Pr(y_i > 250 \mid x_i) =$$

$$\frac{\exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}{1 + \exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}$$



Basis Functions Setup

Polynomial and piecewise-constant regression models are special cases of a *basis function* approach.

$$y_i = \beta_0 + \beta_1 \underbrace{b_1(x_i)} + \beta_2 \underbrace{b_2(x_i)} + \cdots + \beta_K \underbrace{b_K(x_i)} + \varepsilon_i$$

Polynomial $b_1(x) = x$

$b_2(x) = x^2$

$\vdots$

$b_d(x) = x^d$

Step fcn $b_1(x) = C_1(x)$

$b_2(x) = C_2(x)$

$\vdots$

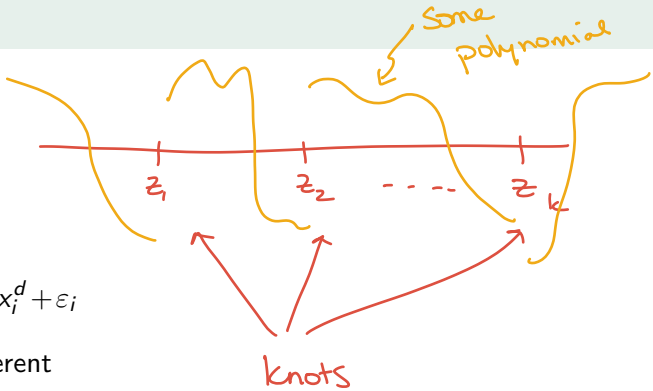
$b_K(x) = C_K(x)$

Section 2

Regression Splines

Some function $f(x)$ to predict y

Piecewise polynomials

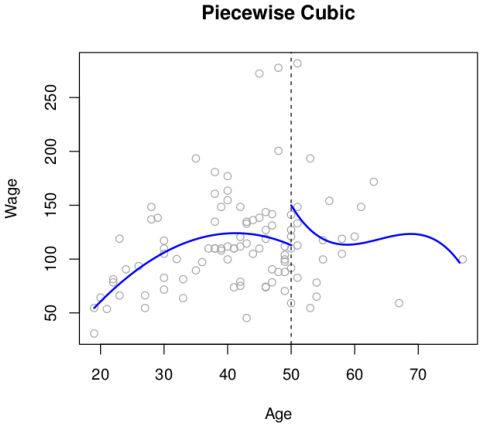


- Fit a polynomial regression

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$

- Let the β_i's be different at different locations of the range.

Example of piecewise polynomial



Example:

deg 3 polynomial = cubic

$$y_i = \begin{cases} \beta_{01} + \beta_{11}x_i + \beta_{21}x_i^2 + \beta_{31}x_i^3 + \epsilon_i & \text{if } x_i < c \\ \beta_{02} + \beta_{12}x_i + \beta_{22}x_i^2 + \beta_{32}x_i^3 + \epsilon_i & \text{if } x_i \geq c. \end{cases}$$

learn 8 variables to get this piecewise polynomial

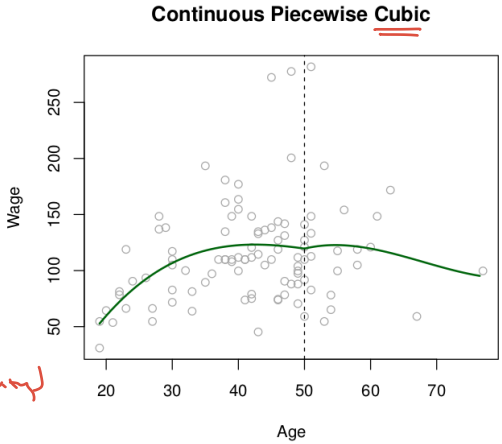
The fix

- Fit piecewise polynomial
- Require continuity at knots

8 input variables

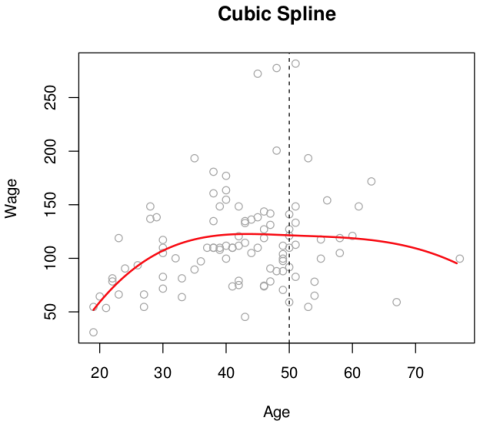
− 1 constraint (force continuity)

= 7 degrees of freedom



The better fix: Cubic splines

- Fit piecewise cubic polynomial
- Require continuity at knots
- Require the first and second derivatives to be continuous at knots



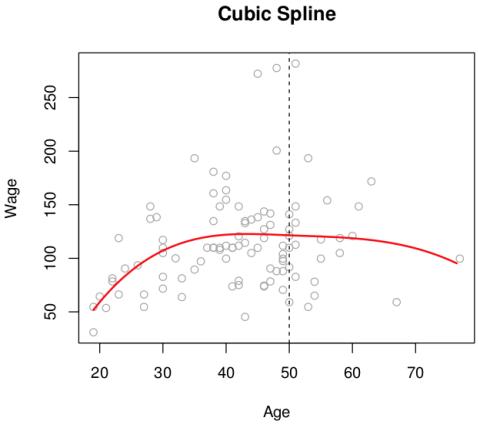
8 variables

−3 constraints (cont, 1st deriv cont, 2nd deriv cont)

5 degrees of freedom

Cubic splines: degrees of freedom

$$f(x) = \begin{cases} \beta_0^1 + \beta_1^1 x + \beta_2^1 x^2 + \beta_3^1 x^3 & x < c \\ \beta_0^2 + \beta_1^2 x + \beta_2^2 x^2 + \beta_3^2 x^3 & x > c \end{cases}$$



Cubic spline w/ K knots:

4 + K degrees of freedom

Spline basis representation

Want to pick b_i so that we represent a cubic spline with K knots as

$$y_i = \beta_0 + \beta_1 \underbrace{b_1(x_i)} + \beta_2 \underbrace{b_2(x_i)} + \cdots + \beta_{K+3} \underbrace{b_{K+3}(x_i)} + \varepsilon_i$$

3+K functions

Truncated power basis function

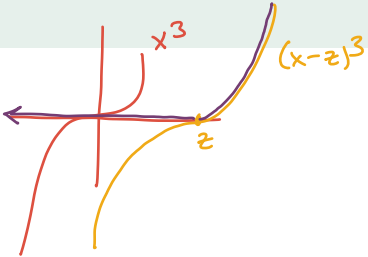
variable

fixed knot

ξ

$$h(x,z) = (x-z)_+^3 = \begin{cases} (x-z)^3 & \text{if } x > z \\ 0 & \textit{else} \end{cases}$$

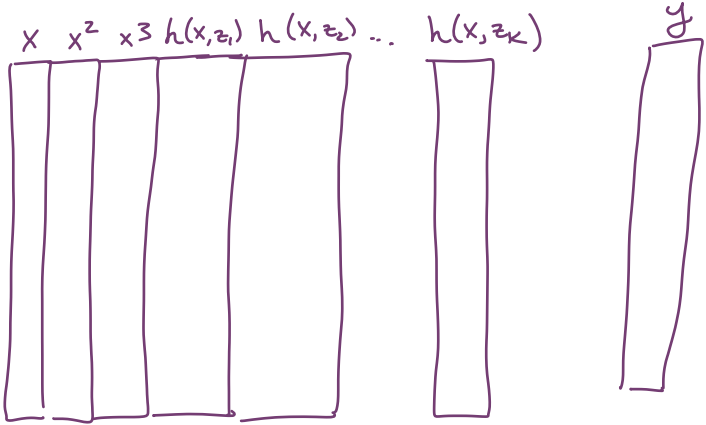
Desmos link: https://www.desmos.com/calculator/esucuulbgj



The basis for cubic splines

Given knots at $z_1, \cdots, z_K$

- X
- X^2
- X^3
- $h(X, z_1)$
- $h(X, z_2)$
- $\vdots$
- $h(X, z_K)$



$$f(X) = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 h(X, z_1) + \beta_5 h(X, z_2) + \cdots + \beta_{k+3} h(X, z_K)$$

Coding example

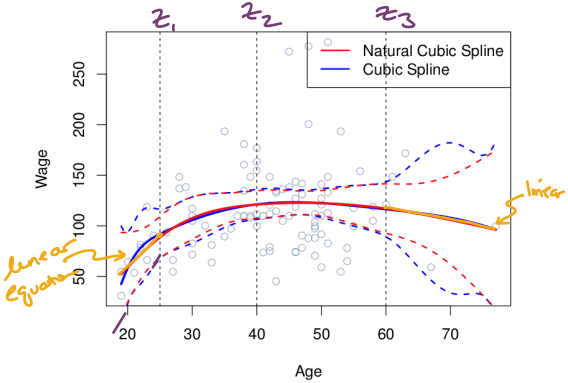
Notes on cubic splines

* Human eyes can't detect where the knots are given cubic spline... reasonable option for "smooth" function

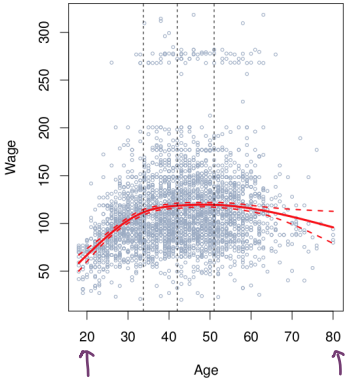
* High variance outside range of input values

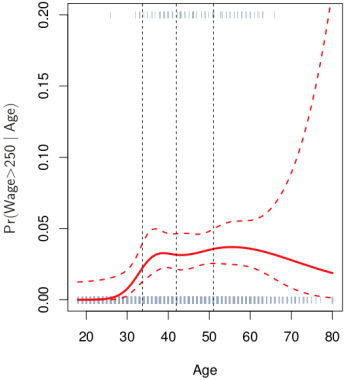
* Natural spline:

  regression spline but require boundary constraints where the function is linear outside input values



Where to put the knots?





Plan A:

Uniform placement inside input range (sklearn default)

Plan B:

Data driven

Put knots @ the quantiles of the data

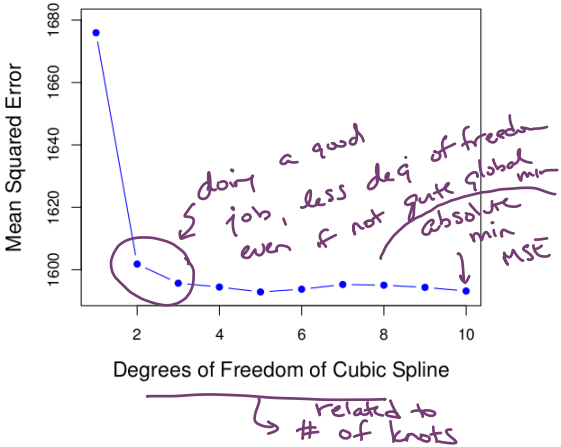
How many knots to use?

When in doubt, Cross-Validate.

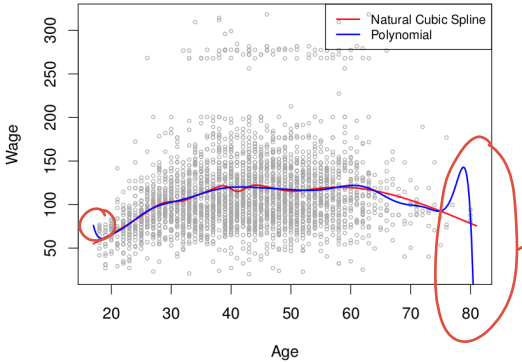
for each # of knots

K-fold CV to get MSE

pick "the best"



Cubic splines vs Polynomial Regression



This example

Cubic spline w/ 15 deg freedom

deg 15 polynomial

wiggly results on end

Splines give more flexibility via knots rather than increasing degree

$\Rightarrow$ Smoother model

Next time

Status	Lec #	Date			Reading	Homeworks
		Mon	Oct 23	No class - Fall break		
		Wed	Oct 25	**Midterm #2**		
Done	20	Fri	Oct 27	Dimension Reduction	6.3	
Done	21	Mon	Oct 30	More dimension reduction; High dimensions	6.4	
Done	22	Wed	Nov 1	Polynomial & Step Functions	7.1,7.2	
Pushed	23	Fri	Nov 3	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
	24	Mon	Nov 6	Regression Splines	7.4	~~HW #6 Due~~
	25	Wed	Nov 8	Decision Trees	8.1	HW #6 Due
	26	Fri	Nov 10	Random Forests	8.2.1, 8.2.2	
	27	Mon	Nov 13	Maximal Margin Classifier	9.1	
	28	Wed	Nov 15	SVC	9.2	
	29	Fri	Nov 17	SVM	9.3, 9.4	
	30	Mon	Nov 20	Single layer NN	10.1	
	31	Wed	Nov 22	Virtual: Project office hours		
		Fri	Nov 24	No class - Thanksgiving		
		Mon	Nov 27	**Review**		
		Wed	Nov 29	**Midterm #3**		
	32	Fri	Dec 1	Multi Layer NN	10.2	
	33	Mon	Dec 4	CNN	10.3	
	34	Wed	Dec 6	Unsupervised Learning & Clustering	12.1, 12.4	
	35	Fri	Dec 8	Virtual: Project office hours		Project due